Office hours Exam

Wednesday, January 22, 2025

Quadratic Term $x \cdot (Ax) = \underbrace{[x_1 \dots x_n]}_{x^T} \underbrace{\begin{bmatrix} a_{11} & & \\ & \ddots & \\ & & a_{nn} \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}}_{x}$ (1)

5. Let $A \in \mathbb{R}^{n\times n}$ be a symmetric positive definite matrix and let $b \in \mathbb{R}^n$. Consider the function

$$f : \mathbb{R}^n \to \mathbb{R},\ f(x) = \frac{1}{2}x^T Ax - b^T x.$$

$\longrightarrow$ Linear Term $b \cdot x = b_1x_1 + \dots + b_nx_n$ (2)

(a) Prove that f has a unique minimum, which satisfies the equation $Ax = b$. **1p**

(b) Write a gradient descent method for finding the minimum of f. **1p**

- Method 1: We compute $\frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_n}$ using (1) + (2).

Then $\nabla f = 0$.

$$f(x) = \frac{1}{2} x^T Ax - b^T x = \frac{1}{2}[x_1 \dots x_n] \begin{bmatrix} a_{11} & \dots & a_{1n} \\ & \ddots & \\ a_{n1} & \dots & a_{nn} \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} - \sum_{i=1}^{n} b_i x_i$$

$$= \frac{1}{2}[x_1 \dots x_n] \begin{bmatrix} \sum_{j=1}^{n} a_{1j} x_j \\ \vdots \\ \sum_{j=1}^{n} a_{nj} x_j \end{bmatrix} - \sum_{i=1}^{n} b_i x_i$$

$$f(x) = \frac{1}{2} \sum_{i=1}^{n} \sum_{j=1}^{n} a_{ij} x_i x_j - \sum_{i=1}^{n} b_i x_i.$$

- Depending on x_1: $\frac{1}{2} \sum_{j=1}^{n} a_{1j} x_1 x_j + \frac{1}{2} \sum_{i=1}^{n} a_{i1} x_i x_1 - b_1 x_1.$

Differentiate in x_1: $a_{11}x_1 + \frac{1}{2}\sum_{j=2}^{n} a_{1j} x_j + \frac{1}{2}\sum_{i=2}^{2} a_{i1} x_i - b_1.$

$a_{1j} = a_{j1}$ (sym)

$$\frac{\partial f}{\partial x_1} = \sum_{j=1}^{n} a_{1j} x_j - b_1, \longrightarrow (Ax-b)_1$$

$$\frac{\partial f}{\partial x_n} = \sum_{j=1}^{n} a_{nj} x_j - b_n \longrightarrow (Ax-b)_n.$$

$$\rightarrow \nabla f(x) = Ax - b.$$

- Method 2: Use Taylor expansion

$$f(x+h) = \underbrace{f(x)}_{\text{Const in } h} + \underbrace{\boxed{\nabla f(x)} \cdot h}_{\text{linear in } h} + \underbrace{\frac{1}{2} h^T \boxed{H(x)} h}_{\text{Quadratic in } h} + \dots$$

$f(x) = \frac{1}{2} x^T Ax - b^T x$, $\quad f(x+h) = \frac{1}{2}(x+h)^T A(x+h) - b^T(x+h).$

$f(x) = \frac{1}{2} x^T A x - b^T x, \quad f(x+h) = \frac{1}{2}(x+h)^T A (x+h) - b^T(x+h).$

$\Rightarrow f(x+h) = \frac{1}{2} x^T A x + \frac{1}{2} x^T A h + \frac{1}{2} h^T A x + \frac{1}{2} h^T A h - b^T x - b^T h.$

$= \underbrace{\frac{1}{2} x^T A x - b^T x}_{= f(x) \checkmark} + \underbrace{\frac{1}{2} \boxed{x^T A h} + \frac{1}{2} \boxed{h^T A x} - b^T h}_{\boxed{(Ax-b)} \cdot h \;\; (Ax-b)^T h} + \underbrace{\frac{1}{2} h^T \boxed{A} h}_{H(x) = A.}$

$x^T A h \overset{?}{=} h^T A x$

- $\boxed{x^T A h} = x \cdot (Ah) = (Ah) \cdot x = (Ah)^T x = h^T A^T x \overset{A^T = A}{=} \boxed{h^T A x.}$
  $= h \cdot (Ax).$

Conclusion:
$$\boxed{\begin{array}{l} f(x) = \frac{1}{2} x^T A x - b^T x. \\ \nabla f(x) = Ax - b, \quad H(x) = A. \end{array}}$$

- $\nabla f(x) = 0 \Longleftrightarrow Ax = b.$ There exists a unique $x \in \mathbb{R}^n$ because A is invertible.
  Fact: A sym. pos. def $\Longrightarrow$ eigenvalues > 0
  $\Longrightarrow \det A =$ product eigenvalues $> 0.$

Gradient descent: $\boxed{x_{n+1} = x_n - s \nabla f(x_n)} = x_n - s(Ax_n - b).$

step size.

Direction of steepest descent: $-\nabla f(x_n).$

6. Let the probabilities $p_1, p_2, p_3 \in (0,1)$ with $\boxed{p_1 + p_2 + p_3 = 1.}$ Consider the function → constraint

$$f : \mathbb{R}^3 \to \mathbb{R},\; f(p_1, p_2, p_3) = -\sum_{i=1}^{3} p_i \log_2(p_i),$$

known as information entropy (a measure of uncertainty for the probability distribution).

(a) Using Lagrange multipliers, find p_1, p_2, p_3 that maximize the entropy function f. **0.75p**

(b) Generalize to n probabilities $p_1, \ldots, p_n$. **0.25p**

CROSS ENTROPY.

$$L(p_1, p_2, p_3, \lambda) = -\sum_{i=1}^{3} p_i \log_2(p_i) + \lambda(p_1 + p_2 + p_3 - 1).$$

$\nabla L = 0: \quad \frac{\partial L}{\partial p_1} = \frac{\partial L}{\partial p_2} = \frac{\partial L}{\partial p_3} = \frac{\partial L}{\partial \lambda} = 0. \Longrightarrow p_1 = p_2 = p_3 = \frac{1}{3}.$

- Critical point: $\nabla f(x) = 0.$ → Local Max, → Local Min, → Saddle
- Extremum = Max or Min.

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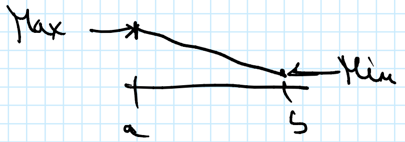
Local Min

Saddle Point.

Fermat: Local Extremum $x \Longrightarrow \nabla f(x) = 0$.



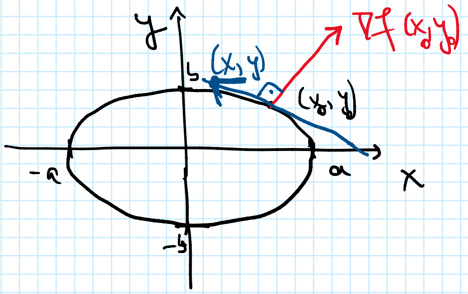
$\underbrace{f(x,y)}=1$

7. Consider the ellipse with equation $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, with $a,b>0$.

(a) Find the equation of the tangent line to the ellipse at a point (x_0, y_0). **1p**

(b) Find the area enclosed by the ellipse, for example by using a double integral. **1p**

$\nabla f \perp$ Level Set



Level Set: $f(x,y)=1$.

$\nabla f(x_0,y_0) \perp (x-x_0, y-y_0)$.

$\nabla f(x_0,y_0)\cdot(x-x_0, y-y_0)=0$.

$f(x,y)=\frac{x^2}{a^2}+\frac{y^2}{b^2}$.

$\left(\frac{\partial f}{\partial x}(x_0,y_0), \frac{\partial f}{\partial y}(x_0,y_0)\right)\cdot(x-x_0, y-y_0)=0$.

$\left(\frac{2x_0}{a^2}, \frac{2y_0}{b^2}\right)\cdot(x-x_0, y-y_0)=0$.

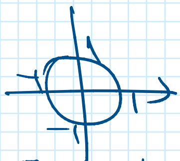
$\Rightarrow \frac{\cancel{2}x_0}{a^2}(x-x_0)+\frac{\cancel{2}y_0}{b^2}(y-y_0)=0$, $\quad \frac{x x_0}{a^2}+\frac{y y_0}{b^2}=\frac{x_0^2}{a^2}+\frac{y_0^2}{b^2}=f(x_0,y_0)=1$.

$\rightarrow \boxed{\frac{x x_0}{a^2}+\frac{y y_0}{b^2}=1.}$, $\quad y\cdot\frac{y_0}{b^2}=1-\frac{x_0}{a^2}x$.

(b) Domain $D=\{(x,y)\in\mathbb{R}^2 \mid \frac{x^2}{a^2}+\frac{y^2}{b^2}\le 1\}$.

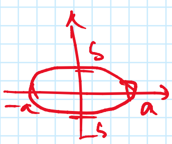
Area of the ellipse $=\iint_D 1\,dx\,dy$.

Domain: $x^2+y^2\le 1$.



Polar coordinates: $x=r\cos\theta$
$y=r\sin\theta$, $r\in[0,1]$, $\theta\in[0,2\pi)$.

Domain: $\frac{x^2}{a^2}+\frac{y^2}{b^2}\le 1$.



$\left(\frac{x}{a}\right)^2+\left(\frac{y}{b}\right)^2=1$.

Modified polar coordinates: $x=a\cdot r\cos\theta$, $\frac{x}{a}=r\cos\theta$

Modified polar coordinates: $x = a \cdot r \cos\theta$, $\frac{x}{a} = r\cos\theta$, $r \in [0,1], \theta \in [0, 2\pi)$.

$y = b \cdot r \sin\theta$, $\frac{y}{b} = r\sin\theta$.

Compute $J = \begin{bmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{bmatrix} = \begin{bmatrix} a\cos\theta & -ar\sin\theta \\ b\sin\theta & br\cos\theta \end{bmatrix}$

$$\det J = abr\cos^2\theta + abr\sin^2\theta = \boxed{abr.}$$

$$\text{Area} = \int_0^1 \int_0^{2\pi} 1 \cdot abr \, d\theta \, dr = ab \cdot \int_0^{2\pi} d\theta \cdot \int_0^1 r\,dr = ab \cdot 2\pi \cdot \frac{1}{2} = ab\pi.$$

4. Let $A = \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix}$ and define the function $f : \mathbb{R}^2 \to \mathbb{R}$, $f(x) = x^T Ax$, for $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$. $\nabla f(x) = 2Ax$

(a) Find the direction of steepest descent for f at the point $(1, 0)$. 0.5p

(b) Find and classify the critical points of f. 1p

(c) Find the minimum of f over the unit circle: $\min_{x\in\mathbb{R}^2} f(x)$ subject to $\|x\|^2 = 1$. 1p

(d) For a symmetric matrix $B \in \mathbb{R}^{n\times n}$, find $\max_{x\in\mathbb{R}^n} x^T Bx$ subject to $\|x\|^2 = 1$. 0.5p

$\|x\|^2 = x_1^2 + x_2^2$

$\partial_{x_1}: 2x_1$, $\partial_{x_2} = 2x_2$

$\nabla g(x) = 2(x_1, x_2) = 2x.$

c) $L(x, \lambda) = \underbrace{x^T A x}_{f(x)} - \lambda \underbrace{(\|x\|^2 - 1)}_{g(x)}.$

$\nabla L = 0$, $\nabla L = \nabla f - \lambda \nabla g(x) = 2Ax - 2\lambda x = 0. \; | \div 2.$

$\boxed{Ax - \lambda x = 0,}$ $(A - \lambda I_n)x = 0 \Rightarrow \lambda$ - eigenvalue of the matrix.

$\boxed{Ax = \lambda x} \Rightarrow f(x) = x^T Ax = x^T \lambda x = \lambda \underbrace{\|x\|^2}_{=1} = \lambda.$

Minimum of f over unit circle = Smallest eigenvalue of A.

5. (a) Compute $\int_0^\infty e^{-x} x^n \, dx$, for $n \in \mathbb{N}$. 1p

$\frac{x^n}{e^x} \xrightarrow{x\to\infty} 0$

$$I(n) = \int_0^\infty e^{-x} x^n dx = -\int_0^\infty (e^{-x})' x^n dx = \underbrace{-e^{-x} x^n \Big|_0^\infty}_{=0} + \int_0^\infty e^{-x} (x^n)' dx$$

$$= n \int_0^\infty e^{-x} x^{n-1} dx.$$

$$I(n) = n\, I(n-1) = n(n-1) I(n-2).$$

$$= n(n-1) \cdots 2 \cdot 1 \cdot \underbrace{I(0)}_{=1} = n!$$

5. Let A be an $m \times n$ matrix, b a vector in R^m and the least squares minimization problem

$$\min_{x \in \mathbb{R}^n} \|Ax - b\|^2.$$

Prove that the solution x^* of this problem satisfies (the so-called normal equations)

$$A^T A x^* = A^T b.$$

$$f(x) = \|Ax-b\|^2 = (Ax-b)\cdot(Ax-b) = (Ax-b)^T(Ax-b)$$

$$= (Ax)^T Ax - \underbrace{(Ax)^T b - b^T Ax}_{=} + b^T b.$$

$$= \underbrace{x^T \boxed{A^T A} x}_{\text{quadratic}} - \underbrace{\boxed{2\, A^T b} \cdot x}_{\text{linear}} + b^T b.$$

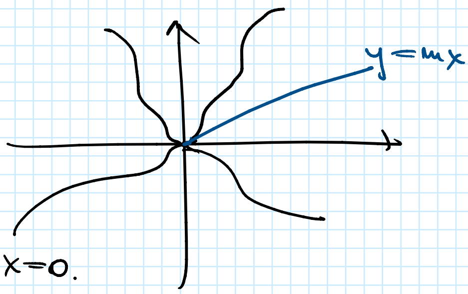
$a \cdot b = a^T b = a_1 b_1 + \dots a_n b_n$

$(AB)^T = B^T A^T.$

• $\nabla f(x) = 0 \iff 2A^TAx - 2A^Tb = 0 \iff A^TAx = A^Tb$

• $b^TAx = b \cdot Ax = Ax \cdot b = (Ax)^Tb = (A^Tb)^T x = x \cdot A^Tb.$

$\shortparallel \ (A^Tb)^T x = A^Tb \cdot x$



Ex: $\displaystyle\lim_{(x,y)\to(0,0)} \frac{x^3+y^3}{x^2+y^2} = 0$

$y = mx$: $\displaystyle\lim_{x\to 0} \frac{x^3+m^3x^3}{x^2+m^2x^2} = \lim_{x\to 0} \frac{1+m^3}{1+m^2} \cdot x = 0.$

• The limit could be 0!

For $x, y > 0$,

$x^3+y^3 \le \underbrace{(x^2+y^2)(x+y)}_{x^3+y^3+x^2y+y^2x}$

$(*)$ $\quad 0 \le \left| \dfrac{x^3+y^3}{x^2+y^2} \right| \le |x|+|y|.$

$\downarrow$ 0 (obviously!)

$$|x^3+y^3| \overset{?}{\le} (x^2+y^2)(|x|+|y|).$$

$$|x^3+y^3| \le |x|^3+|y|^3 \le (x^2+y^2)(|x|+|y|). \quad \text{TRUE} \iff \text{TRUE}.$$

8. Compute the following integrals:

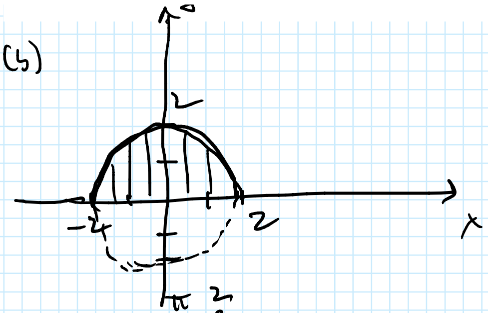
(a) $\displaystyle\int_0^1 \int_x^1 e^{-y^2}\, dy\, dx.$ 1p

(b) $I = \displaystyle\iint_D \sin(x^2+y^2)\, dx\, dy$, where $D = \{(x,y) \in \mathbb{R}^2 \mid \underbrace{x^2+y^2 \le 4}_{\text{disk}},\ y \ge 0\}$. 1p

(b)

$x^2+y^2=4 \rightarrow$ Circle of radius 2.

(b)



$x^2+y^2=4 \rightarrow$ Circle of radius 2.

<u>Polar coordinates</u>: $x = r\cos\theta \quad r\in[0,2]$

$x^2+y^2=r^2.$ $\quad y = r\sin\theta, \quad \theta\in[0,\pi].$

$\det J = r.$

$$I = \int_0^{\pi}\int_0^{2} \sin(r^2)\cdot r \, dr\, d\theta = \underbrace{\left(\int_0^{\pi} d\theta\right)}_{=\pi}\left(\int_0^2 r\sin(r^2)\,dr\right) = \pi\cdot\left[-\frac{1}{2}\cos(r^2)\Big|_0^2\right]$$

$$= \pi\left(-\frac{1}{2}\cos 4 + \frac{1}{2}\right).$$

5. (a) Compute $\int_0^{\infty} e^{-x}x^n \, dx$, for $n\in\mathbb{N}$. 1p

(b) Compute $\iint_R \max\{x,y\}\, dx\, dy$, where $R=[0,1]\times[0,1]$. 1p

$$\max\{x,y\} = \begin{cases} x, & x\geq y \\ y, & x<y \end{cases} \qquad x:\ [0,1] = [0,y]\cup(y,1].$$

$$\iint_R \max\{x,y\}\, dx\, dy = \int_0^1\left[\int_0^y \underbrace{y}_{\max\{x,y\}}\, dx + \int_y^1 \underbrace{x}_{\max\{x,y\}}\, dx\right] dy$$

$$= \int_0^1 \left(y^2 + \frac{1-y^2}{2}\right) dy = \frac{1}{3} + \frac{1}{2} - \frac{1}{6} = \frac{4}{6} = \frac{2}{3}.$$